Erdem Bıyık

https://ebiyik.github.io
https://scholar.google.com/citations?user=P-G3sjYAAAAJa

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Current Position

- **University of Southern California** — Los Angeles, CA, USA
  Assistant Professor of Computer Science — *Aug 2023 – present*
  Assistant Professor of Electrical and Computer Engineering - Systems (by courtesy) — *Jan 2024 – present*

Education

- **Stanford University** — Stanford, CA, USA
  Doctor of Philosophy in Electrical Engineering; GPA: 4.07/4.00 — *Sep 2017 – Jun 2022*
  Advisor: Prof. Dorsa Sadigh
- **Stanford University** — Stanford, CA, USA
  Master of Science in Electrical Engineering; GPA: 4.08/4.00 — *Sep 2017 – Apr 2019*
- **Bilkent University** — Ankara, Türkiye
  Bachelor of Science in Electrical and Electronics Engineering; GPA: 4.00/4.00 — *Aug 2012 – Jun 2017*
  Rank: 1/170 based on GPA
- **National University of Singapore** — Singapore
  Exchange Student in Electrical and Computer Engineering — *Aug 2015 – Dec 2015*

Work Experience

- **UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI)** — Berkeley, CA, USA
  Postdoctoral Researcher — *Jul 2022 - Jul 2023*
  - **Reward Learning**: Working on various active reward learning projects for robotics with Prof. Stuart Russell and Prof. Anca Dragan
  - **Reinforcement Learning for Deep Brain Stimulation**: Working on the applications of reinforcement learning algorithms for deep brain stimulation under the supervision of Prof. Anca Dragan
- **Google Research** — Mountain View, CA, USA
  Research Intern — *Jun 2021 - Sep 2021*
  - **Recommender Systems**: Working on active preference elicitation for recommender systems under the supervision of Dr. Yinlam Chow, Dr. Mohammad Ghavamzadeh, and Prof. Craig Boutilier
  - **Preference-based Reinforcement Learning**: Working on relaxing initial state assumptions in preference-based reinforcement learning under the supervision of Dr. Yinlam Chow and Dr. Mohammad Ghavamzadeh
- **National Magnetic Resonance Research Center (UMRAM)** — Ankara, Türkiye
  Undergraduate Researcher & Research Intern — *Apr 2016 - Sep 2017*
  - **csMRI**: Developing compressed sensing methods for accelerated MRI under the supervision of Prof. Tolga Çukur
  - **SSFP Imaging**: Developing coil compression methods and artifact suppression techniques for balanced SSFP under the supervision of Prof. Tolga Çukur
- **ASELSAN** — Ankara, Türkiye
  Intern (2015), Research Engineer (2017) — *Jun 2015 - Jul 2015, Apr 2017 - Aug 2017*
  - **Military Communications (2015)**: Developing a C++ program to decode and encode data in MIL-STD-3014 protocol with an easy-to-use interface
  - **Algorithms Design (2015)**: Designing algorithms to solve composite launch acceptability region problem

- **SSFP Imaging (2017)**: R&D projects related to field inhomogeneity, banding profile estimation, coil compression and banding suppression in bSSFP MRI under the supervision of Dr. Aykut Koç

- **École polytechnique fédérale de Lausanne (EPFL)** — Lausanne, Switzerland
  Research Intern via Summer@EPFL Program — *Jun 2016 - Sep 2016*
  - **Approximate Message Passing**: Research about approximate message passing algorithms as an intern in the Information Processing Group under the supervision of Prof. Rüdiger Urbanke

- **Anadolu Agency** — Ankara, Türkiye
  Intern — *Jun 2014 - Jul 2014*
  - **Image Processing**: Developing a Java, C++ and MySQL based program that uses image registration, matching and comparison techniques to detect copyright infringements

Teaching Experience

- **University of Southern California** — Los Angeles, CA
  Instructor — *Fall 2023, Spring 2024, Fall 2024*
  - **CSCI445L Introduction to Robotics (Spring 2024)**: Teaching the course
  - **CSCI699 Robot Learning (Fall 2023, Fall 2024)**: Designing and teaching the course

- **Stanford University** — Stanford, CA
  Teaching Assistant — *Winter 2020, Winter 2021*
  - **CS237B / EE260B / AA174B / AA274B Principles of Robot Autonomy II**: Holding office hours and sections, preparing and grading homeworks and exams
    Awarded "outstanding course assistant" by the CS department (top 5%) in Winter 2021

- **Bilkent University** — Ankara, Türkiye
  Teaching Assistant — *Spring 2014, Fall 2016*
  - **CS114 Introduction to Programming for Engineers (2014)**: Teaching in the weekly tutorials & recitations
  - **EEE211 Analog Electronics (2016)**: Teaching, interviewing and grading students in the laboratory sessions

- **Stanford University** — Stanford, CA
  Guest Lecturer — *Winter 2022, Winter 2025*
  - **CS333 Safe and Interactive Robotics (2022)**: Teaching a lecture on "learning from human preferences"
  - **CS237B / EE260B / AA174B / AA274B Principles of Robot Autonomy II (2025)**: Giving a talk on "Efficient Robot Learning via Interaction with Humans"

Mentoring

- **Current Ph.D. Students**: Jesse Zhang (co-advised with Jesse Thomason and Joseph Lim), Anthony Liang (co-advised with Stephen Tu), Pavel Czempin, Yigit Korkmaz, Yutai Zhou, Yimin Tang (co-advised with Sven Koenig), Ayano Hiranaka (co-advised with Daniel Seita)
- **Current M.Sc. Students**: Xinhu Li, Dhanush Kumar Varma Penmetsa, Zhaojing Yang, Matthew Hong, Yusen Luo
- **Current Undergraduate Students**: Jaiv Doshi, Urvi Bhuwania, Ashley Lan, Kevin Kim
- **Ph.D. Alumni**
  - Ayush Jain (2024, co-advised with Joseph Lim). Next: Research Scientist at Meta.
  - Sumedh Sontakke (2024, co-advised with Laurent Itti). Next: Applied Scientist at Amazon Robotics.
- **M.Sc. Alumni**
  - Thomas Reeves (2024). Next: Ph.D. Student in Computer Science at USC.

Honors & Awards

- **2024 Outstanding Paper Finalist at TMLR**: Awarded to 5 papers in 2024. Awarded for the paper "Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback"
- **Stanford CS Outstanding Course Assistant**: Awarded to the top 5% of CAs in the department. Awarded for the "Principles of Robot Autonomy II" course in Winter 2021
- **Qualcomm Innovation Fellowship 2020 North America**: Finalist (as a group of two PhD students)
- **HRI 2020 Honorable Mention in Technical Advances**: Awarded to 3 papers in the Technical Advances track. Awarded for the paper "When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans"
- **Qualcomm Innovation Fellowship 2019 North America**: Finalist (as a group of two PhD students)
- **Stanford University James D. Plummer Graduate Fellowship (2017-2018)**: Full tuition waiver & stipend during the first year of PhD program
- **Bilkent University Electrical and Electronics Engineering Department, Graduation Awards (2017)**: Academic Excellence, Research Excellence, Voluntary Professional Activities, Social Awareness and Activities
- **Scholarship of the Turkish Prime Ministry (2012-2017)**: Awarded monthly stipend during the BSc program (given to those who rank in first 100 among 1.8 million students in nationwide university entrance exam)
- **Bilkent University Comprehensive Scholarship (2012-2017)**: Full tuition waiver & stipend during the BSc program
- **IEEExtreme Programming Competitions**: 2^{nd} in Türkiye, 78^{th} among all participants, 2015. 1^{st} in Türkiye, 73^{rd} among all participants, 2014. 1^{st} in Türkiye, 100^{th} among all participants, 2013 (All as a group of three students)
- **Turkish Intelligence Foundation (TZV) Marathon**: Ranked twice in top 25 and once 6^{th}, 2012-2014
- **İşbank Golden Youth Award (2012)**: Granted for outstanding performance in the nationwide university entrance exam
- **Nationwide University Entrance Exam (LYS)**: Ranked 13^{th} among 1.8 million students in Türkiye, 2012

Invited Talks & Poster Presentations

- Asking Easy Questions: A User-Friendly Approach to Active Reward Learning
  - Inzva & AI Safety Türkiye, AI Safety Talks (2025)
  - CoRL 2019 (poster)
- Robot Learning with Minimal Human Feedback
  - Stanford University, Guest Lecture at "Principles of Robot Autonomy II" (2025)
  - University of Pennsylvania, General Robotics, Automation, Sensing, & Perception Lab (GRASP) Seminar (2025)
  - UC Santa Barbara, The Center for Control, Dynamical Systems, and Computation Seminar (2025)
  - Bilkent University, Electrical and Electronics Engineering Department, Graduate Seminar (2024)
  - University of Washington, Robotics Seminar (2024)
  - University of Illinois Urbana-Champaign, Human-Centered Autonomy Lab (2024)
- Open Problems in Reinforcement Learning from Human Feedback and Potential Solutions for Data-Efficiency
  - Caltech AI Alignment Club (2025)
  - University of Amsterdam, The Amsterdam Machine Learning Lab (2023)
- Efficient Robot Learning via Interaction with Humans
  - New Faculty Highlights at the 39th Annual AAAI Conference on Artificial Intelligence, 2025
  - Southern California Robotics (SCR) Symposium 2023
- Reinforcement Learning from Comparative Language Feedback

  - The Inaugural Conference of the International Association for Safe and Ethical AI (IASEAI 2025 – poster)

- Trajectory Improvement and Reward Learning from Comparative Language Feedback
  - CoRL 2024 (poster)

- Preference Learning from Minimal Human Feedback for Interactive Autonomy
  - ODSC West 2024 Conference (Open Data Science Conference)
  - International Conference on Micro, Nano and Smart Systems (IC-MNSS) 2024
  - IIT Hyderabad, Mechanical and Aerospace Engineering, Department Seminar & Technology Innovation Hub on Autonomous Navigation and Data Acquisition Systems (2024)

- Making Robot Learning More Natural Through Human Saliency and Language
  - RSS 2024 Workshop on "Mechanisms for Mapping Human Input to Robots: From Robot Learning to Shared Control/Autonomy"

- Data- and Time-Efficient Learning of Human Preferences
  - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2024)

- Optimizing Robot Behavior via Comparative Language Feedback
  - Human-Interactive Robot Learning (Workshop at HRI 2024 – poster)

- Data-Efficient Learning from Human Feedback and Large Pretrained Models for Robotics
  - Arizona State University, School of Computing and Augmented Intelligence (2023)

- ViSaRL: Visual Reinforcement Learning Guided by Human Saliency
  - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2023 – poster)

- Learning Preferences for Interactive Autonomy
  - Middle East Technical University, Robotics and AI Technologies Application and Research Center (2022)
  - Sonoma State University, Engineering Colloquium (2022)
  - Cornell University, Robotics Seminar (2022)
  - University of Wisconsin-Madison, Computer Science, Department Seminar (2022)
  - University of Illinois Urbana-Champaign, Computer Science, Department Seminar (2022)
  - University of Michigan, Robotics Institute, Department Seminar (2022)
  - University of Southern California, Computer Science, Department Seminar (2022)
  - Imperial College London, Department of Aeronautics, Aerodynamics & Control Seminar (2022)
  - Carnegie Mellon University, Robotics Institute, Department Seminar (2022)
  - UCLA, Electrical and Computer Engineering, Department Seminar (2022)
  - Bilkent University, Electrical and Electronics Engineering, Graduate Seminar (2021)
  - UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI), Beneficial AI Seminar (2021)
  - Sabancı University, Computer Science & Engineering, Department Seminar (2021)
  - Koç University, AI Meetings (2021)
  - Bay Area Robotics Symposium (BARS) 2021 (poster)
  - Virginia Tech, Mechanical Engineering, Department Seminar (2021)
  - Caltech Yue Lab (2021)
  - UT Austin, Personal Autonomous Robotics Lab (2021)
  - UC Berkeley, InterACT Lab (2021)

- Learning Multimodal Rewards from Rankings
  - 6th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2022)
  - CoRL 2021
- Learning from Humans for Adaptive Interaction
  - HRI Pioneers 2022
- APReL: A Library for Active Preference-based Reward Learning Algorithms
  - Human-Interactive Robot Learning (HIRL) (Workshop at HRI 2022)
  - HRI 2022
  - AI-HRI 2021 at AAAI Fall Symposium Series
- Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams
  - AAAI 2022
  - AI-HRI 2021 at AAAI Fall Symposium Series
- The Role of Representations in Human-Aware Learning and Control *(with Dorsa Sadigh)*
  - "Aware-Learning: How to Benefit from Priors" Workshop @ CDC 2021
- Interactive Robotics through the Lens of Learning *(with Dorsa Sadigh)*
  - NCCR (The National Centre of Competence in Research, Switzerland) Automation Seminar, 2021
- Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction
  - Center for Automotive Research at Stanford (CARS) Annual Meeting (2021 – poster)
- Learning Reward Functions from Scale Feedback
  - CoRL 2021 (poster)
- Learning how to Dynamically Route Autonomous Vehicles on Shared Roads
  - ETH Zurich, Institute for Dynamic Systems and Control, Autonomy Talks (2021)
  - The 32nd IEEE Intelligent Vehicles Symposium Workshop (2021)
  - 3rd NorCal Control Workshop (2021)
  - Stanford Robotics Lunch (2019)
  - Center for Automotive Research at Stanford (CARS) Annual Meeting (2020 – poster)
- Walking the Boundary of Learning and Interaction *(with Dorsa Sadigh)*
  - 3rd Robot Learning Workshop: Grounding Machine Learning Development in the Real World @ NeurIPS 2020
- When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans *(with Minae Kwon)*
  - University of Chicago, Graduate Seminar on Topics in Human-Robot Interaction (2020)
- Active Preference-Based Gaussian Process Regression for Reward Learning
  - RSS 2020
- The Green Choice: Learning and Influencing Human Decisions on Shared Roads
  - CDC 2019
- Active Learning of Reward Dynamics from Hierarchical Queries
  - IROS 2019

- Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models
  - ACC 2019
  - Stanford AI Safety Retreat 2019 (poster)

- Batch Active Preference-Based Learning of Reward Functions
  - CoRL 2018
  - Stanford HAI 2019 (poster)
  - TRI Joint University Workshop 2019 (poster)
  - Stanford SystemX Fall 2018 (poster)
  - BARS - Bay Area Robotics Symposium (BARS) 2018 (poster)

Professional Activities

- **Conferences Organized**
  - **Local Arrangements Co-Chair**, Robotics: Science and Systems (RSS), 2025
  - **Organizer**, Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting (Workshop at Robotics: Science and Systems (RSS)), 2025
  - **Finance Chair**, Conference on Robot Learning (CoRL), 2024
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2021
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2020
  - **Organizer**, Emergent Behaviors in Human-Robot Systems (Workshop at Robotics: Science and Systems (RSS)), 2020
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2018

- **Program Committee Member**
  - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025
  - The 39th Annual AAAI Conference on Artificial Intelligence, AI Alignment Track (2025)
  - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
  - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2023

- **Editorial Board**
  - **Associate Editor** for Robot Learning, IEEE Robotics and Automation Letters (RA-L), 2023 – present
  - **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
  - **Area Chair** for Robotics: Science and Systems - Pioneers Workshop 2025
  - **Area Chair** for Late Breaking Results at the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025
  - **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
  - **Area Chair**, Responsible AI (RAI) (Workshop at International Conference on Learning Representations (ICLR)), 2021

- **Panelist**
  - Human Value Learning Panel at the 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024

- **Reviewer for Journals**

- ACM Transactions on Autonomous and Adaptive Systems (TAAS)
- ACM Transactions on Human-Robot Interaction (THRI)
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
- Artificial Intelligence
- Autonomous Agents and Multi-Agent Systems
- Autonomous Robots (AURO)
- Cognitive Systems Research
- Frontiers in Robotics and AI
- Nature Communications
- IEEE Control System Letters (L-CSS)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Robotics and Automation Magazine (RAM)
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Automation Science and Engineering (T-ASE)
- IEEE Transactions on Cognitive and Developmental Systems (TCDS)
- IEEE Transactions on Control Systems Technology (TCST)
- IEEE Transactions on Human-Machine Systems (THMS)
- IEEE Transactions on Intelligent Transportation Systems (T-ITS)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMC-S)
- International Journal of Machine Learning and Cybernetics
- Journal of Machine Learning Research (JMLR)
- MDPI Robotics
- Proceedings of the Royal Society A
- The International Journal of Robotics Research (IJRR)

- **Reviewer for Conferences**
  - IEEE Conference on Decision and Control (CDC), 2019–2020, 2023–2025
  - Robotics: Science and Systems (RSS), 2020–2021, 2023–2025
  - Conference on Uncertainty in Artificial Intelligence (UAI), 2023–2025
  - International Conference on Machine Learning (ICML), 2025
  - International Conference on Learning Representations (ICLR), 2025
  - International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
  - Conference on Neural Information Processing Systems (NeurIPS), 2023–2024
  - Conference on Robot Learning (CoRL), 2019–2024
  - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, 2022–2024
  - IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), 2024
  - Learning for Dynamics & Control (L4DC), 2020, 2023–2024
  - IEEE International Conference on Robotics and Automation (ICRA), 2021–2024

- ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2020–2024
- International Joint Conference on Artificial Intelligence (IJCAI), 2023
- American Control Conference (ACC), 2019, 2021–2022
- IEEE Conference on Control Technology and Applications (CCTA), 2021
- IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2020
- International Conference on Computer-Aided Verification (CAV), 2019
- ACM International Conference on Hybrid Systems: Computation and Control (HSCC), 2019

- **Reviewer for Workshops**
  - Workshop on User-Aligned Assessment of Adaptive AI Systems (AIA) at IJCAI 2025
  - RSS Pioneers Workshop 2023–2025
  - HRI Pioneers Workshop 2023–2024
  - Interactive Learning with Implicit Human Feedback Workshop (ILHF) at ICML 2023
  - AAMAS Workshop on Rebellious and Disobedient AI (RaD-AI), 2022–2023
  - CoRL Workshop on Aligning Robot Representations with Humans, 2022
  - NeurIPS Workshop on Foundation Models for Decision Making (FMDM), 2022
  - NeurIPS Workshop on Progress and Challenges in Building Trustworthy Embodied AI (TEA), 2022
  - NeurIPS Workshop on Machine Learning Safety, 2022
  - RSS Workshop on Learning from Diverse, Offline Data (L-DOD), 2022
  - RSS Workshop on Social Intelligence in Humans and Robots, 2022
  - Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, 2021
  - ICRA Workshop on Social Intelligence in Humans and Robots, 2021
  - Bridging AI and Cognitive Science (BAICS) (Workshop at International Conference on Learning Representations (ICLR)), 2020

Committee Membership for Other Students

- **University of Southern California, Computer Science**
  - Bingjie Tang, Thesis Proposal (May 2024), Dissertation Defense (May 2025)
  - Kegan Strawn, Qualifying Exam (Oct. 2024), Thesis Proposal (May 2025)
  - Bryon Tjanaka, Thesis Proposal (Apr. 2025)
  - Shihan Zhao, Qualifying Exam (Apr. 2025)
  - Soumyaroop Nandi, Thesis Proposal (Apr. 2025)
  - Robby Costales, Qualifying Exam (Sep. 2024), Thesis Proposal (Apr. 2025)
  - Grace Zhang, Qualifying Exam (May 2024), Thesis Proposal (Apr. 2025)
  - I-Chun Liu, Qualifying Exam (Mar. 2025)
  - Anand Balakrishnan, Thesis Proposal (Dec. 2024)
  - Sophie Hsu, Qualifying Exam (Dec. 2024)
  - Kiran Lekkala, Dissertation Defense (Nov. 2024)
  - Zhonghao Shi, Qualifying Exam (Jun. 2024)
  - Xin Zhu, Thesis Proposal (Jun. 2024)
  - Sheryl Paul, Qualifying Exam (May 2024)
  - Yang Chen, Qualifying Exam (Mar. 2024)

- Weizhe Chen, Qualifying Exam (Mar. 2024)
- Varun Sreedhara Bhatt, Qualifying Exam (Feb. 2024)
- Jeremy Morgan, Qualifying Exam (Nov. 2023)

- **University of Southern California, Electrical and Computer Engineering**
  - Javier Borquez, Dissertation Defense (Jun. 2025)
  - Yuning Zhang, Qualifying Exam (May 2025)
  - Zeming Cheng, Qualifying Exam (Sep. 2024), Dissertation Defense (Apr. 2025)
  - Mustafa Altay Karamuftuoglu, Qualifying Exam (May 2024), Dissertation Defense (Apr. 2025)
  - Mingye Li, Qualifying Exam (Jan. 2025)

Outreach and Mentoring

- **USC Thomas Lord Department of Computer Science, 2024-present** Los Angeles, CA, USA
  - **Undergraduate Mentoring Program Organizer**: Organizing the program where PhD students mentor undergraduates about their future career, getting involved in CS research, and applying for industry or graduate school

- **USC Robotics and Autonomous Systems Center (RASC), 2024-present** Los Angeles, CA, USA
  - **Faculty Mentor for USC Robotics Seminar (URoS)**: Mentoring Ph.D. students in running a biweekly robotics seminar to host students presentations and invited speakers
  - **Faculty Mentor for Robotics and Autonomous Systems Center (RASC) Blog**: Mentoring Ph.D. students in running a blog website about the robotics and autonomous systems resarch at USC

- **USC Viterbi School of Engineering, K-12 STEM Center, 2025** Los Angeles, CA, USA
  - **Instructor at CS@SC Coding Camps**: Teaching middle school and high school students "how robots learn to imitate humans" in two sessions

- **Berkeley Artificial Intelligence Safety Initiative for Students, 2023-2024** Berkeley, CA, USA
  - **Mentor at Supervised Program for Alignment Research**: Mentoring three undergraduate students on a multi-agent inverse reinforcement learning research project

- **Berkeley Artificial Intelligence Research Lab (BAIR), 2022-2023** Berkeley, CA, USA
  - **BAIR Undergraduate Mentoring Program Mentor**: Mentoring undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school

- **Stanford University Department of Computer Science, 2020-2022** Stanford, CA, USA
  - **CS Mentorship Program Organizer**: Organizing the CS mentorship program and the related social events
  - **CS Mentorship Program Mentor**: Mentoring first and second-year undergraduate students about their future career, getting involved in research projects, and applying for industry or graduate school

- **Stanford University AI Laboratory (SAIL), 2018-2020** Stanford, CA, USA
  - **AI Mentorship Program Organizer**: Organizing the AI mentorship program and the related social events
  - **AI Mentorship Program Mentor**: Mentoring first and second-year undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
  - **Robotics Lunch Organizer (2019-2020)**: Organizing bi-weekly robotics lunch sessions where invited professors, postdoctoral researchers and Ph.D. candidates present their research

- **Stanford STEM to SHTEM Summer Internship Program, 2019** Stanford, CA, USA
  - **Mentor**: Mentoring high school students in a research project on decision making under risk / time constraints

- **IEEE Bilkent Student Branch, 2012-2017** Ankara, Türkiye

- **Road to University Volunteer (2013-2017)**: Introducing engineering and campus life to high school students from all around Türkiye
- **Vice Chair (2014-2015)**: Managing and organizing the technical trainings and social events
- **Robotics and Automation Society Member (2012-2015)**: Assistantship in the electronics and robotics focused tutorials
- **Web Team Member (2012-2014) and Webmaster (2013-2014)**: Designing and managing the website of IEEE Bilkent SB, its communities and events; teaching in web design tutorials and in MATLAB tutorials; organizing a Java programming competition and weekly brain teasers

- **GazeteBilkent, 2014-2017** Ankara, Türkiye
  - **Online Operations Manager**: Managing the website of GazeteBilkent, an online newspaper

- **Bilkent University, 2014-2017** Ankara, Türkiye
  - **Webmaster of Electrical and Electronics Engineering (2013-2017), Mechanical Engineering (2015-2017), Economics (2016-2017) Departments**: Designing, developing and managing the websites of the departments and developing necessary web tools
  - **1st and 2nd Industrial Design Projects Fairs Student Coordinator and Webmaster (2015, 2016)**: Organizing the fair in which the senior students of Bilkent EEE present their industry projects
  - **Graduate Research Conference '15 Student Coordinator (2015)**: Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
  - **Graduate Research Conference '14 Webmaster and Organization Team Member (2014)**: Organizing the conference series in which graduate students of Bilkent EEE present their research and projects

- **Bilkent Chess Society, 2013-2014** Ankara, Türkiye
  - **Vice Chair**: Organizing tournaments on campus, participating in local events as a team

- **Bilkent Academic Career Club, 2013** Ankara, Türkiye
  - **Physics Olympiads Volunteer**: Organizing a physics competition among high school students

Patents

1. Z Cao, **E Bıyık**, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh. "Reinforcement Learning Based Control of Imitative Policies for Autonomous Driving", US Patent Application No. US17/002,650.

Journal Publications

1. K Lekkala, H Bao, SA Sontakke, **E Bıyık**, L Itti. "Value Explicit Pretraining for Learning Transferable Representations", IEEE Robotics and Automation Letters (RA-L), 2025. *(Submitted)*
2. K Baraka, I Idrees, TK Faulkner, **E Bıyık**, S Booth, M Chetouani, DH Grollman, A Saran, E Senft, S Tulli, A Vollmer, A Andriella, H Beierling, T Horter, J Kober, I Sheidlower, ME Taylor, SV Waveren, X Xiao. "Human-Interactive Robot Learning: Definition, Challenges, and Recommendations", ACM Transactions on Human Robot Interaction (THRI), 2024. *(Submitted)*
3. **E Bıyık**, N Anari, D Sadigh. "Batch Active Learning of Reward Functions from Human Preferences", ACM Transactions on Human-Robot Interaction (THRI), 2024.
4. S Casper*, X Davies*, C Shi, TK Gilbert, J Scheurer, J Rando, R Freedman, T Korbak, D Lindner, P Freire, T Wang, S Marks, CR Segerie, M Carroll, A Peng, P Christoffersen, M Damani, S Slocum, U Anwar, A Siththaranjan, M Nadeau, EJ Michaud, J Pfau, D Krasheninnikov, X Chen, L Langosco, P Hase, **E Bıyık**, A Dragan, D Krueger, D Sadigh, D Hadfield-Menell (*equal contribution). "Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback", Transactions on Machine Learning Research (TMLR), 2023. ***Outstanding paper finalist.***
5. **E Bıyık**, N Huynh, MJ Kochenderfer, D Sadigh. "Active Preference-Based Gaussian Process Regression for Reward Learning and Optimization", The International Journal of Robotics Research (IJRR), 2023; doi: 10.1177/02783649231208729.
6. M Tucker, K Li, E Novoseller, M Pétriaux, G Burger, **E Bıyık**, M Masselin, D Sadigh, JW Burdick, Y Yue, AD Ames. *Anonymous Submission*, Nature Machine Intelligence, 2022. *(Submitted)*

7. **E Bıyık**, DP Losey, M Palan, NC Landolfi, G Shevchuk, D Sadigh. "Learning Reward Functions from Diverse Sources of Human Feedback: Optimally Integrating Demonstrations and Preferences", The International Journal of Robotics Research (IJRR), 2022; doi:10.1177/02783649211041652
8. DA Lazar*, **E Bıyık***, D Sadigh, R Pedarsani (*equal contribution). "Learning How to Dynamically Route Autonomous Vehicles on Shared Roads", Transportation Research Part C: Emerging Technologies (TR_C), 2021; doi: 10.1016/j.trc.2021.103258
9. **E Bıyık***, DA Lazar*, R Pedarsani, D Sadigh (*equal contribution). "Incentivizing Efficient Equilibria in Traffic Networks with Mixed Autonomy", IEEE Transactions on Control of Network Systems (TCNS), 2021; doi: 10.1109/TCNS.2021.3084045.
10. **E Biyik***, K Keskin*, SUH Dar, A Koc, T Çukur (*equal contribution). "Factorized sensitivity estimation for artifact suppression in phase-cycled bSSFP MRI", NMR in Biomedicine, 2020; doi: 10.1002/nbm.4228.
11. **E Biyik**, E Ilicak, T Çukur. "Reconstruction by Calibration over Tensors for Multi-Coil Multi-Acquisition Balanced SSFP Imaging", Magnetic Resonance in Medicine (MRM), 2017; doi: 10.1002/mrm.26902.
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14. Y Ling*, K Owalekar*, O Adesanya, **E Bıyık**, D Seita (*equal contribution). "IMPACT: Intelligent Motion Planning with Acceptable Contact Trajectories via Vision-Language Models", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, Oct. 2025. *(Submitted)*
15. Y Tang*, X Xiong*, J Xi, J Li, **E Bıyık**, S Koenig (*equal contribution). "RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, Oct. 2025. *(Submitted)*
16. X Li, A Jain, Z Yang, Y Korkmaz, **E Bıyık**. *Anonymous Submission*, 9th Conference on Robot Learning (CoRL), Seoul, South Korea, Sep. 2025. *(Submitted)*
17. M Hong*, A Liang*, KJ Kim, HB Rajaprakash, J Thomason†, E Bıyık†, J Zhang† (*equal contribution, †equal advising). *Anonymous Submission*, 9th Conference on Robot Learning (CoRL), Seoul, South Korea, Sep. 2025. *(Submitted)*
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20. A Jain, N Kosaka, X Li, K Kim, **E Bıyık**, J Lim. *Anonymous Submission*, 2nd Reinforcement Learning Conference (RLC), Edmonton, Alberta, Canada, Aug. 2025. *(Submitted)*
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26. Z Yang, M Jun, J Tien, SJ Russell, A Dragan, **E Bıyık**. "Trajectory Improvement and Reward Learning from Comparative Language Feedback", 8th Conference on Robot Learning (CoRL), Munich, Germany, Nov. 2024.
27. J Zhang, M Heo, Z Liu, **E Bıyık**, J Lim, Y Liu, R Fakoor. "EXTRACT: Efficient Policy Learning by Extracting Transferable Robot Skills from Offline Data", 8th Conference on Robot Learning (CoRL), Munich, Germany, Nov. 2024.

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